

The Non-Rigid Rotor

When greater accuracy is desired, the departure of the molecular rotational spectrum from that of the `rigid_rotor` model can be described in terms of centrifugal distortion and the vibration-rotation interaction. To second order in the relevant quantum numbers, the rotation can be described by the expression

$$E_{v,J} = \left[hc\omega_e(v + \frac{1}{2}) - hc\omega_e x_e(v + \frac{1}{2})^2 \right]$$

vibration terms

$$+ \frac{hcB_e J(J+1) - hcD_e J^2(J+1)}{\text{rotation term}}$$

$$-hc\alpha_e(v + \frac{1}{2})J(J+1)$$

vibration

$$\begin{array}{c} \vdots \\ r_e \end{array} \quad \begin{array}{c} \vdots \\ r_e \end{array}$$

 Δ

Rigid rotor term Centrifugal stretching term

Centrifugal stretching
term

 Δ $\sqrt{r^2}$

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R Nerve

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